

Finding the Sweet Spot: Open Source Academia AI

Finding the Sweet Spot: Open Source Academia AI

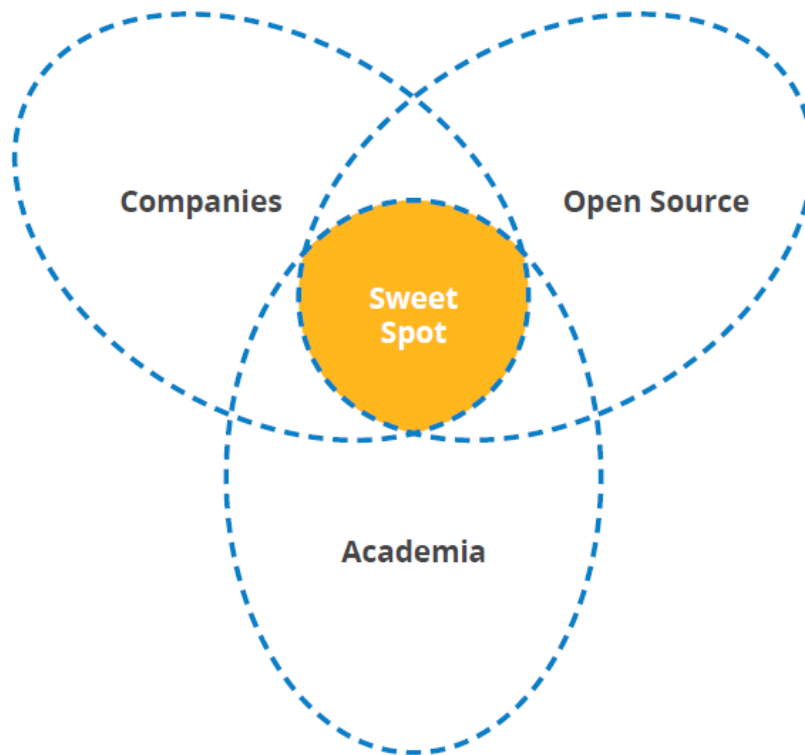
Created on 2018-05-24 16:06

Published on 2018-05-24 16:31

The following is adapted from "Open Source AI: Projects, Insights and Trends" published by The Linux Foundation (free e-book download); a guide to 16 open source AI projects that examines their history, contributions trends, and insights gained from studying their development statistics.

The overnight academic success with AI research is the result of decades of hard work, government funding, and tireless professors and students who kept on going despite the various challenges. Open source creates a great environment for the academic world to collaborate with the industry without imposing too many restrictions on that type of relationship.

The sweet spot is the intersection in interest from the academia, the open source community and enterprises. In recent years, interest in AI started to grow incredibly fast in enterprises at a time when AI has been an established R&D domain in Academia with reference implementations to many of the ideas already existing as open source projects. The interests aligned and today the rest is history.



Finding the sweet spot – $\text{Companies} \cap \text{Open Source} \cap \text{Academia}$

For instance, the appetite from companies to open up AI R&D labs in Canada or capture existing academic labs under their sponsorship in cities like Montreal, Toronto, and Vancouver, is driven by the eagerness to capture the abundance of Canadian AI talent. This talent is the result of decades of government and tax payers support in that domain combined with the use of open source as a tool to collaborate with other academics on the implementation side.

Many similarities are worth noting between Academia and open source:

- Their collaborative approach to innovation
- The process of creating and validating new ideas
- The licenses applied to the source code resulting from the R&D
- Community building – although they build communities in different circles, they do share the passion and practice
- The importance of sharing results at conferences and getting external reviews and feedback

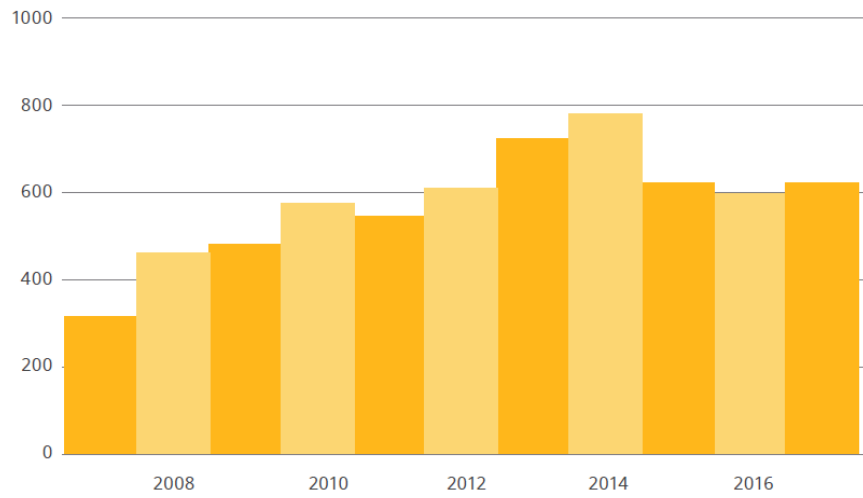
In the AI field, companies joined the party and they were able to put in an incredible amount of resources into the space and improve on the general Academic approach and practices in areas such as:

- The finality of produced software
- The approach to architecture (plugins, APIs)
- Accelerating the inception of initiatives
- Building a developer ecosystem around these projects
- Gathering contributions for the projects from other companies (typically business partners)
- Providing access to large data sets that can be used for training specific models
- Improving documentation, developer tools, and developer support in general
- Increasing API coverage and support, and
- Applying these technologies to products and services

The result was the culmination of efforts between Academia and enterprises under an open source umbrella that created the right conditions for such a collaboration to happen. Many of the AI related open source project, platform, framework and libraries started as academic R&D initiated at different universities.

For this study, we collaborate with Teqmine to mine over 12.5 million issued patents in the US and the EU. We found that academic institutions were granted 8,597 AI patents for the period extending from January 1, 2006 to December 31, 2017. The top 10 academic patentees sorted by highest number of issued patents are:

1. California University System
2. Massachusetts Institute of Technology
3. Columbia University
4. California Institute of Technology
5. Johns Hopkins University
6. Stanford University
7. University of Michigan
8. University of Texas
9. Hong Kong University of Science and Technology
10. New York University



Number of issued AI patents to Academic institutions (2006-2017)

The very interesting part is the realization that some of these academic institutions have generated more AI patents than many enterprises that fall under the Fortune 100 technology companies. We will explore this data in a follow up publication where we examine the AI patent landscape.

Download the free e-book! _____